

Adversarial Search: Summary and Other Games

Introduction

For decades, games have served as a crucial proving ground for Artificial Intelligence research, pushing the boundaries of what machines can achieve. From the strategic depths of chess to the complex probabilities of poker, these adversarial environments offer a structured yet challenging domain for developing and testing intelligent algorithms. This lecture explores the historical milestones of AI in various games, the underlying search algorithms that enabled these victories, and the classification of games based on information and chance. Understanding how AI conquers these complex challenges provides insight into the fundamental principles of search, decision-making under uncertainty, and the continuous pursuit of optimal, or at least highly effective, strategies in dynamic environments. The journey through game AI reveals not only the impressive capabilities of machines but also highlights the evolving partnership between human and artificial intelligence, pointing towards a future where collaboration, rather than pure competition, defines the cutting edge.

Core Concepts

Term	Definition	Significance
Deterministic Games	Games where the outcome of any action is fully predictable and there is no element of randomness.	These games allow for precise calculation of future states, making algorithms like Minimax highly effective.
Perfect Information	Games where all players have complete knowledge of the game state at all times.	Crucial for algorithms that rely on evaluating all possible moves and their consequences, as seen in Chess.
Minimax Algorithm	A decision-making algorithm used in two-player, zero-sum games with perfect information, aiming to minimize the maximum	Forms the foundation for AI in many classic board games, enabling machines

Term	Definition	Significance
	possible loss for a worst-case scenario.	to play optimally or near-optimally.
Branching Factor	The number of possible moves or actions available from a given game state.	A high branching factor significantly increases the complexity and computational cost of search algorithms, as seen in Go.
Monte Carlo Search	A tree search technique that uses random sampling to estimate the value of nodes in a game tree, particularly useful for games with very high branching factors.	Revolutionized AI's ability to tackle games like Go, where traditional Minimax is computationally infeasible.
Games of Chance	Games that incorporate random elements, such as rolling dice or drawing cards, affecting the game state or outcomes.	Requires algorithms to account for probabilities and expected values, moving beyond purely deterministic evaluations.
Expectiminimax	An extension of the Minimax algorithm designed for games that include both adversarial opponents and elements of chance.	Allows AI to make rational decisions by calculating expected values at chance nodes, optimizing for average outcomes.
Imperfect Information	Games where players do not have complete knowledge of the game state, such as hidden cards in poker.	Introduces significant challenges for AI, requiring reasoning under uncertainty and often involving probabilistic models.

Detailed Analysis

1. Historical Milestones and Algorithmic Advancements in Game AI

The history of Artificial Intelligence is punctuated by significant achievements in game playing, demonstrating the power of computational search and evaluation. These milestones often involved specialized algorithms and massive computational resources.

- **Checkers (1994): Chinook's Triumph**

- In 1994, the Chinook software defeated the human world champion in checkers, ending a 40-year human reign.
- Chinook utilized a pre-computed endgame database, allowing it to perfectly play out the final stages of the game.
- It evaluated an astonishing 444 billion positions, effectively "solving" checkers for practical purposes.

- **Chess (1997): Deep Blue vs. Kasparov**

- IBM's Deep Blue famously defeated Garry Kasparov, the reigning world chess champion, in 1997.
- Deep Blue employed a sophisticated combination of techniques:
 - Opening game databases for initial moves.
 - Endgame databases for final positions.
 - Minimax search in the middle game, often reaching depths of 12 to 14 moves.
 - Undisclosed methods for extending search lines, sometimes reaching depths of up to 40 moves in critical situations.
- The sheer depth of its search capabilities overwhelmed human intuition and calculation. Today, chess software is so advanced that humans rarely compete against machines. Interestingly, the best chess players are often human-machine teams, highlighting a collaborative future.

- **Othello: Machine Dominance**

- In Othello, machines have become so proficient that human champions no longer compete against them, acknowledging the AI's superior play.

- **Go (2016-2017): AlphaGo's Breakthrough**

- Go was considered an "oddity" for a long time because machines were too weak to challenge humans. Its complexity far surpassed chess:
 - Go uses a 19x19 board compared to chess's 8x8.
 - Games typically last 300 moves, versus 100 in chess.

- The average branching factor in Go is 235, significantly higher than chess's 35, making traditional deep search infeasible (typically only 3-4 depth possible).
- DeepMind's AlphaGo, in 2016, defeated Lee Sedol (considered the "Federer of Go").
- AlphaGo's success stemmed from a novel combination of deep neural networks and tree search, specifically **Monte Carlo Search**.
- Instead of exhaustively searching the entire game tree, Monte Carlo Search intelligently samples parts of the subtree, repeating this process to estimate move values.
- In 2017, AlphaGo further defeated Ke Jie, another top Go champion, solidifying AI's dominance in Go.
- **Poker (2015): Heads-Up Limit Hold'em Solved**
- Poker, a game of imperfect information and chance, was considered a much harder challenge.
- In 2015, the University of Alberta demonstrated that at least "heads-up limit Hold'em poker" could be solved.
- This achievement required months of large-scale cluster computation, enabling AI to play as well as, and eventually defeat, human experts.
- **The Next Frontier: Dota**
- With many traditional board games conquered, the AI community is now looking towards more complex, real-time strategy games.
- Games like Dota (Defense of the Ancients), a multiplayer online battle arena (MOBA) game, represent the current frontier due to their highly dynamic environments, imperfect information, and complex decision spaces. OpenAI is actively researching AI for Dota.

2. Classifying Games: Information, Chance, and Determinism

Games can be categorized based on two primary dimensions: whether they involve chance and whether players have perfect information. This classification dictates the type of AI algorithms required.

- **Deterministic Games with Perfect Information:**
- **Characteristics:** No randomness, all players know the complete game state.
- **Examples:** Chess, Checkers, Othello.
- **AI Approach:** Primarily Minimax and its variations, often enhanced with alpha-beta pruning, opening books, and endgame databases.

- **Games of Chance with Perfect Information:**
- **Characteristics:** Random elements (e.g., dice rolls), but all players know the complete game state (including outcomes of random events).
- **Example:** Monopoly.
- **AI Approach:** Requires **Expectiminimax** to account for probabilities at chance nodes.
- **Deterministic Games with Imperfect Information:**
- **Characteristics:** No randomness, but players do not have complete knowledge of the game state (e.g., hidden information).
- **Example:** Certain card games where cards are hidden but no shuffling or drawing occurs during play. (The lecture notes this category but doesn't provide a specific common example, stating "everybody knows. Not everybody knows everything, but the game proceeds deterministically.")
- **AI Approach:** More complex, often involving belief states and reasoning about opponent's knowledge.
- **Games of Chance with Imperfect Information:**
- **Characteristics:** Both random elements (e.g., drawing cards) and hidden information (e.g., opponent's hand).
- **Examples:** Poker, Bridge.
- **AI Approach:** The most challenging category, requiring sophisticated probabilistic reasoning, opponent modeling, and often Monte Carlo methods or extensive pre-computation.

3. Expectiminimax: Handling Chance in Adversarial Search

When games introduce elements of chance, the standard Minimax algorithm is insufficient because it cannot account for probabilistic outcomes. **Expectiminimax** extends Minimax by introducing "chance nodes" into the game tree.

- **Structure of the Game Tree:**
- **Max Nodes:** Represent turns where the AI player makes a choice to maximize its score.
- **Min Nodes:** Represent turns where the opponent makes a choice to minimize the AI's score.
- **Chance Nodes:** Represent points where a random event occurs (e.g., a dice roll), leading to different outcomes with specific probabilities.
- **Evaluation at Chance Nodes:**

- Instead of picking the best or worst outcome, a chance node's value is calculated as the expected value of its possible outcomes.
- This is done by summing the product of each outcome's probability and its value.
- **Example:**
 - Consider a chance node with two possible outcomes:
 - Outcome 1: Value of 5, probability 0.4
 - Outcome 2: Value of 4, probability 0.6
 - The expected value at this chance node would be: $(0.4 * 5) + (0.6 * 4) = 2.0 + 2.4 = 4.4$
- **Integration with Min/Max Nodes:**
 - Max nodes will choose the action that leads to the highest expected value from their children (which could be min or chance nodes).
 - Min nodes will choose the action that leads to the lowest expected value from their children.
- **Time Complexity:** The time complexity of Expectiminimax is $b^m * n^m$, where b is the branching factor, m is the depth, and n is the number of chance outcomes at each chance node. This exponential complexity highlights the computational challenge.

4. The Enduring Lessons of Game Playing for AI

Game playing has provided profound insights into the nature of AI and its limitations.

- **Search Depth and Cutoffs:** Games emphasize the need for deep search, but also the necessity of intelligent cut-off strategies due to computational limits.
- **Evaluation Functions:** What to do at the time of cutoff, i.e., how to evaluate a non-terminal state, is critical for effective play.
- **Variable Search Extension:** Techniques to extend search in a variable fashion, focusing computational effort where it matters most.
- **Meta-Reasoning:** The idea of deciding how much time to spend thinking about a particular move or line of play.
- **Approximation over Optimality:** The realization that perfection is often unattainable, and optimality is a luxury. AI frequently has to approximate and select good-enough solutions that can still outperform the next best alternative. This insight is fundamental to real-world AI applications.

- **Demonstrating AI's Potential:** Traditionally, game-playing programs have been a primary vehicle for AI researchers to demonstrate the field's progress and capabilities to the world. While AI's focus is now broadening to areas like medical diagnosis, games remain a powerful testament to its potential.

Key Takeaways

- [] AI has achieved significant milestones in various games, from checkers and chess to Go and poker, often by leveraging advanced search algorithms and massive computational power.
- [] Games are classified by the presence of chance and the completeness of information, which dictates the appropriate AI approach.
- [] The Minimax algorithm is fundamental for deterministic games with perfect information, aiming to optimize for the worst-case scenario.
- [] Expectiminimax extends Minimax to handle games of chance by calculating expected values at probabilistic nodes.
- [] Monte Carlo Search, which uses intelligent sampling, was crucial for AI's success in games with extremely high branching factors like Go.
- [] The "branching factor" and desired "search depth" are critical determinants of a game's computational complexity for AI.
- [] AI in games often prioritizes effective approximation over unattainable optimality, a key lesson for real-world applications.
- [] The frontier of game AI is moving towards complex, real-time, imperfect information environments like MOBA games (e.g., Dota).
- [] The best players in some games are now human-machine combinations, suggesting a future of collaborative intelligence.

Conclusion

The journey of AI through the landscape of games, from the early triumphs in checkers to the recent mastery of Go and poker, underscores a profound evolution in artificial intelligence capabilities. These successes are not merely about winning; they are about developing sophisticated search algorithms, probabilistic reasoning, and strategic decision-making under varying degrees of uncertainty and information. The transition from deterministic, perfect-information games to those involving chance and hidden information has pushed AI to innovate with techniques like Expectiminimax and Monte Carlo Search, demonstrating adaptability and learning. As AI continues to tackle increasingly complex environments like real-time strategy games, the lessons learned from game playing—particularly the emphasis

on approximation, meta-reasoning, and efficient search—remain invaluable. Ultimately, the narrative of game AI is shifting from pure machine dominance to a more nuanced understanding of human-machine collaboration, hinting at a future where the most formidable intelligence emerges from the synergy between human intuition and artificial computational power, extending its impact far beyond the gaming board into diverse real-world applications.